NB: Open with an advanced pdf reader (e.g., Acrobat to have animations)

Lecture on Conformal Prediction (3/4)

Margaux Zaffran

October 29-30, 2026

LIDAM – ISBA seminar series

FONDATION MATHÉMATIQUE JACQUES HADAMARD

Laboratoire de Mathématiques d'Orsay

LIDAM | Louvain Institute of Data Analysis and Modeling in economics and statistics

ISBA

Previous lectures

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory) (practical session)
3. Towards conditional coverage? (practical session) (theory) (case studies)

Achievable definition of distribution-free validity

$\widehat{C}_\alpha$ = estimated predictive set based on n data points.

Definition (Distribution-free validity)**.**

$\widehat{C}_\alpha$ achieves distribution-free validity if:

- for any distribution $\mathcal{D}$,
- for any associated exchangeable joint distribution $\mathcal{D}^{\text{exch}(n+1)}$,

we have that:

$$\mathbb{P}_{\mathcal{D}^{\text{exch}(n+1)}}\left(Y_{n+1} \in \widehat{C}_\alpha\left(X_{n+1}\right)\right) \geq 1-\alpha.$$

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$\hookrightarrow$ The definition of the conformity scores is crucial, as they incorporate almost all the information: data + underlying model

Split Conformal Prediction: summary

- **Simple** procedure which quantifies the uncertainty of **any** predictive model $\hat{A}$ by returning predictive regions
- **Finite-sample** guarantees
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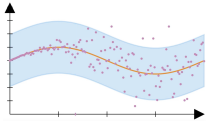
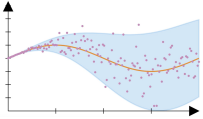
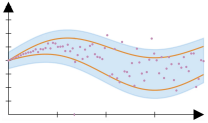
$\hookrightarrow$ marginal also over the whole calibration set and the test point!

SCP: what choices for the regression scores?

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathrm{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S})\}$$

	Standard SCP Vovk et al. (2005)	**Locally weighted SCP** Lei et al. (2018)	**CQR** Romano et al. (2019)
$\mathrm{s}(\hat{A}(X), Y)$	$\lvert\hat{\mu}(X) - Y\rvert$	$\frac{\lvert\hat{\mu}(X) - Y\rvert}{\hat{\rho}(X)}$	$\max(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X) - Y,$ $Y - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X))$
$\widehat{C}_\alpha(x)$	$[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})]$	$[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})\hat{\rho}(x)]$	$[\widehat{\mathrm{QR}}_{\mathrm{lower}}(x) - q_{1-\alpha}(\mathcal{S});$ $\widehat{\mathrm{QR}}_{\mathrm{upper}}(x) + q_{1-\alpha}(\mathcal{S})]$
Visu.			
✓	black-box around a "usable" prediction	black-box around a "usable" prediction	adaptive
✗	not adaptive	limited adaptiveness	no black-box around a "usable" prediction

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4. Beyond exchangeability (methods) (case studies)
5. Computational and statistical trade-offs (methods) (theory)
6. Handling missing data (methods)

Beyond exchangeability

- Some (short) literature review
- Case study: electricity price forecasting

Avoiding data splitting: full conformal and out-of-bags approaches

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- ✗ Label shift, i.e. $\mathcal{L}_Y$ changes but $\mathcal{L}_{X|Y}$ stays constant
- ✗ Arbitrary distribution shift
- ✗ Possibly many shifts, not only one

Covariate shift (Tibshirani et al., 2019)

- Setting:
  - $(X_1, Y_1), \ldots, (X_n, Y_n) \overset{i.i.d.}{\sim} P_X \times P_{Y|X}$
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  1. estimate the likelihood ratio $w(X_i) = \dfrac{\mathrm{d}\tilde{P}_X(X_i)}{\mathrm{d}P_X(X_i)}$

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    $\hookrightarrow$ If the learnt model is accurate and the data noise is strongly mixing, then CP is valid asymptotically ✓
  - Barber et al. (2022)
    $\hookrightarrow$ Quantifies the coverage loss depending on the strength of exchangeability violation
    $$\mathbb{P}(Y_{n+1} \in \widehat{C}_\alpha(X_{n+1})) \geq 1 - \alpha - \underbrace{\text{average violation of exchangeability by each calibration point}}$$
    $\hookrightarrow$ proposed algorithm: reweighting again!
    e.g., in a temporal setting, give higher weights to more recent points.

Online setting

- Data: T_0 random variables $(X_1, Y_1), \ldots, (X_{T_0}, Y_{T_0})$ in $\mathbb{R}^d \times \mathbb{R}$
- Aim: predict the response values as well as predictive intervals for T_1 subsequent observations $X_{T_0+1}, \ldots, X_{T_0+T_1}$ sequentially: at any prediction step $t \in [\![T_0 + 1, T_0 + T_1]\!]$, $Y_{t-T_0}, \ldots, Y_{t-1}$ have been revealed
- Build the smallest interval $\widehat{C}_\alpha^t$ such that:

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$\rightsquigarrow$ More during the case study!

Recent developments (as of late 2023...)

- Consider splitting strategies that respect the temporal structure

Non exhaustive references.

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- Angelopoulos et al. (2023) combines CP ideas with control theory ones, to adaptively improve the predictive intervals depending on the errors structure

Non exhaustive references.

Beyond exchangeability

- Some (short) literature review
- Case study: electricity price forecasting

Avoiding data splitting: full conformal and out-of-bags approaches

Forecasting French spot electricity prices

Hourly day-ahead market prices (between producers and suppliers)

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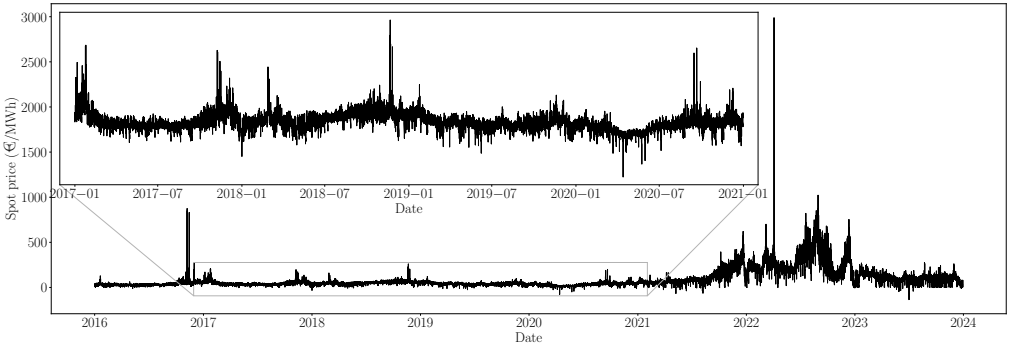

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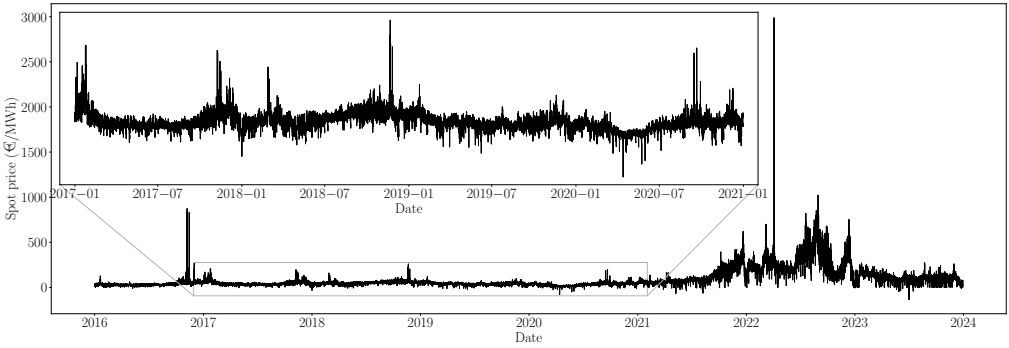
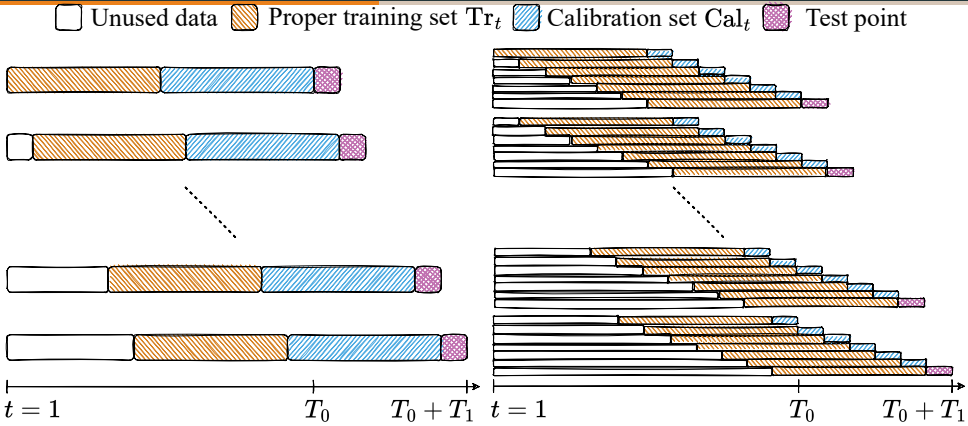


To which extent are they forecastable?

↪ forecasts errors **no lower than 10%** of the realized price!

Temporal splitting strategies: Online Sequential Split Conformal Prediction (OSSCP, Zaffran et al., 2022; Dutot et al., 2024)



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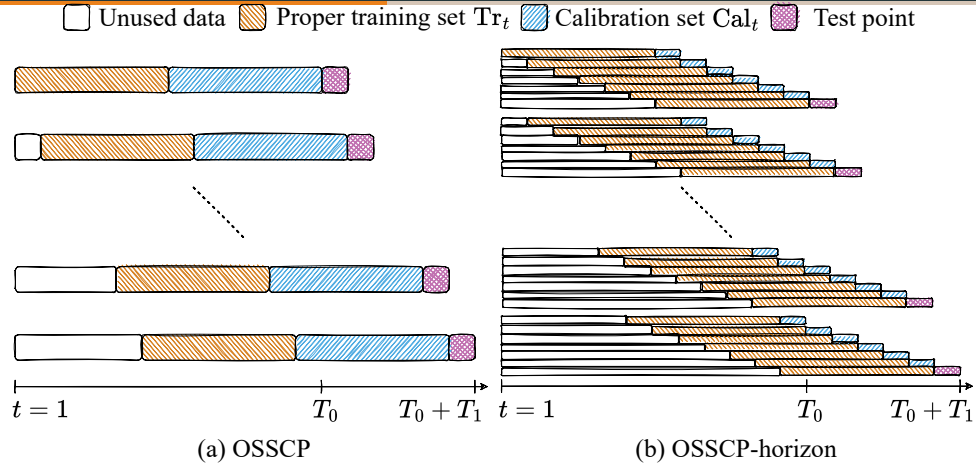

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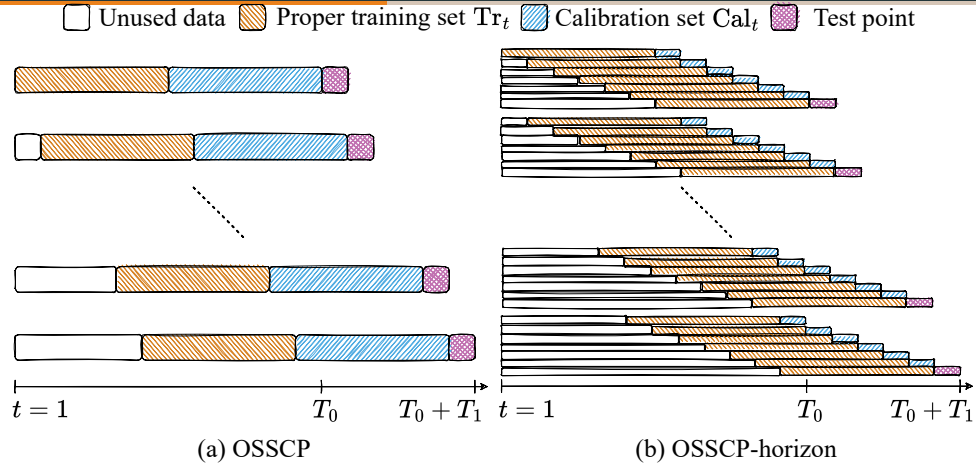

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↪ `OSSCP-horizon` drastically improves robustness in non-stationary temporal settings.

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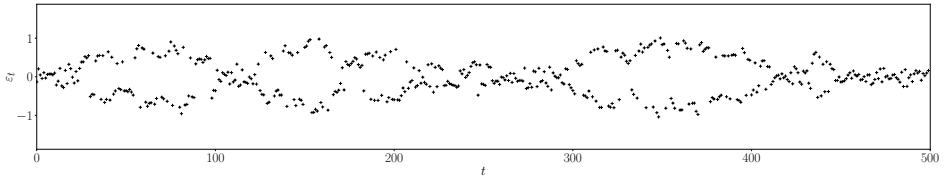
$\Rightarrow$ favors large γ.

Visualisation of ACI procedure



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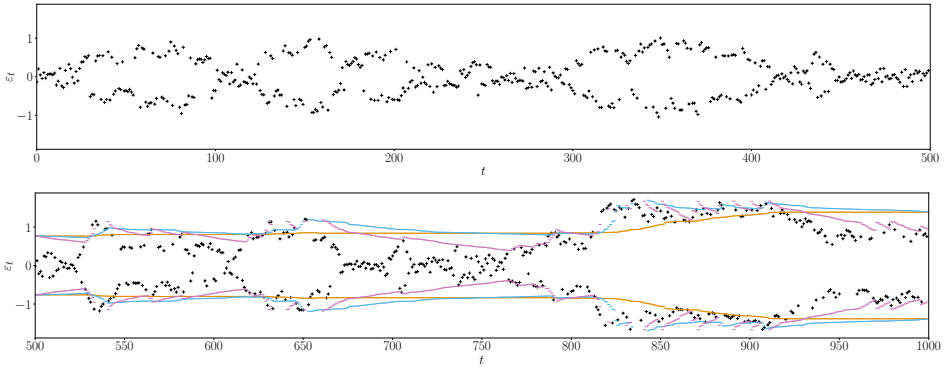
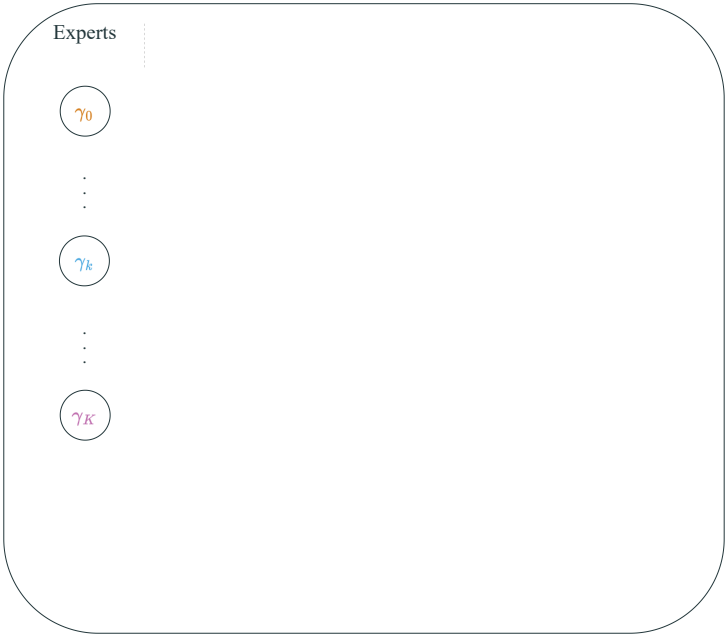


Figure 1: Visualisation of ACI with different values of γ ($\gamma = 0$, $\gamma = 0.01$, $\gamma = 0.05$)

AgACI: adaptive wrapper around ACI, upper bound (Zaffran et al., 2022)



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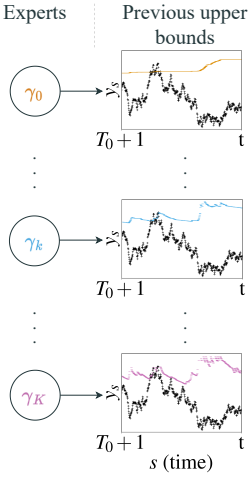

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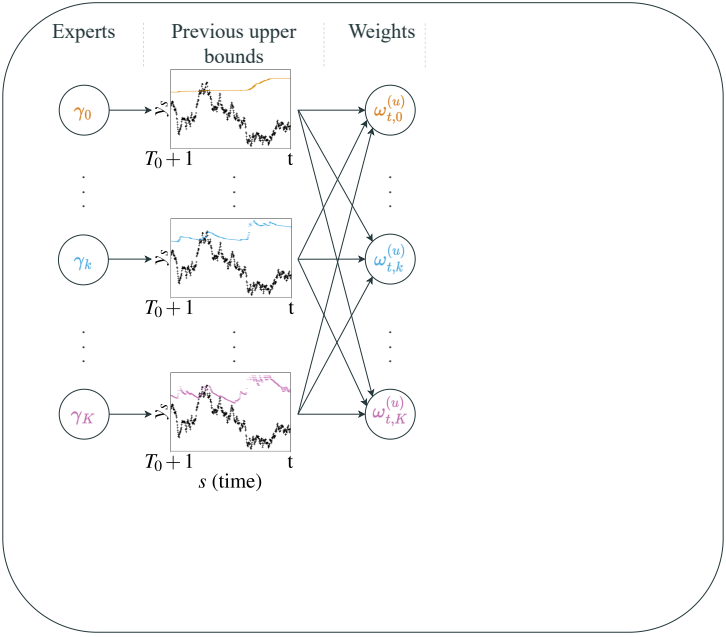

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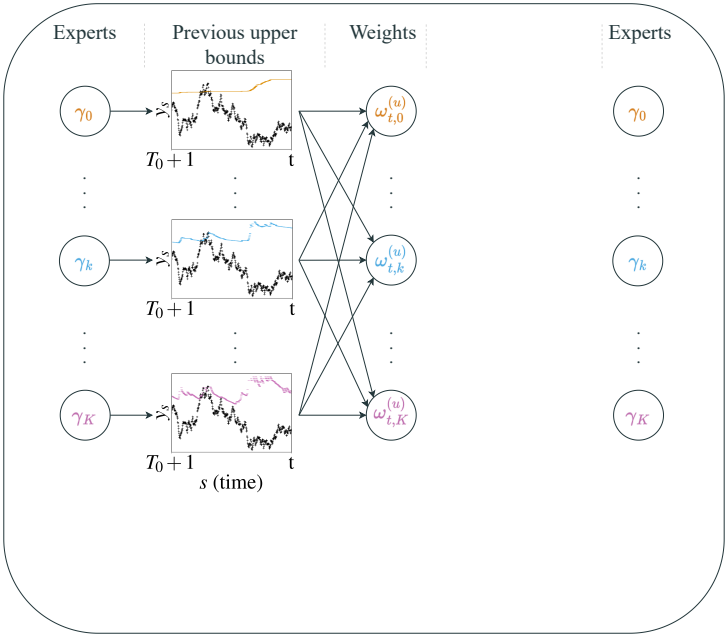

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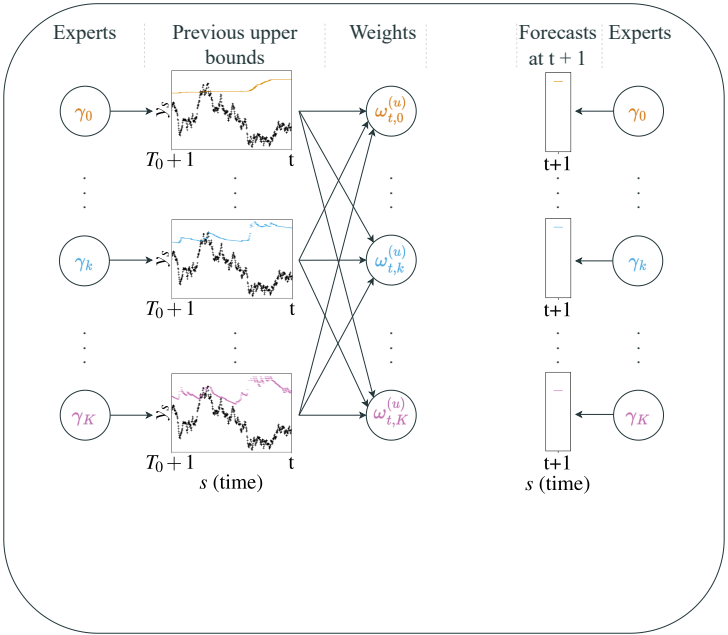

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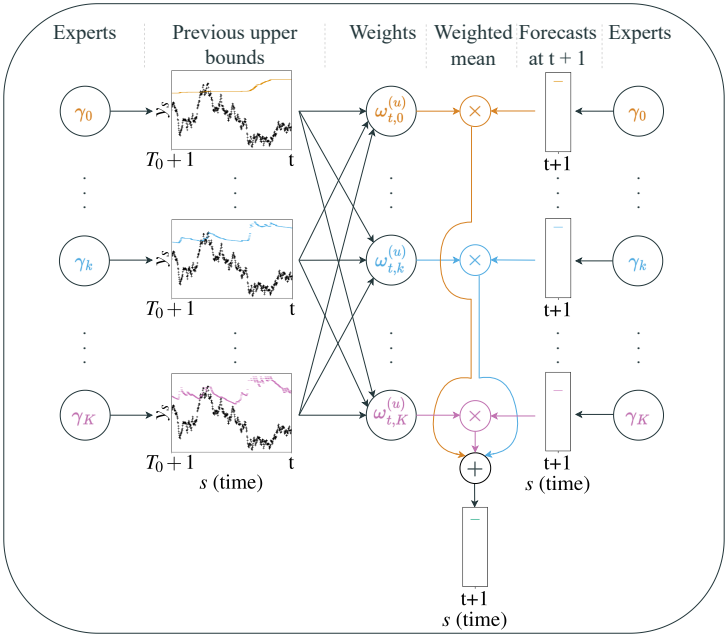

Experimental take-away messages (Zaffran et al., 2022; Dutot et al., 2024)

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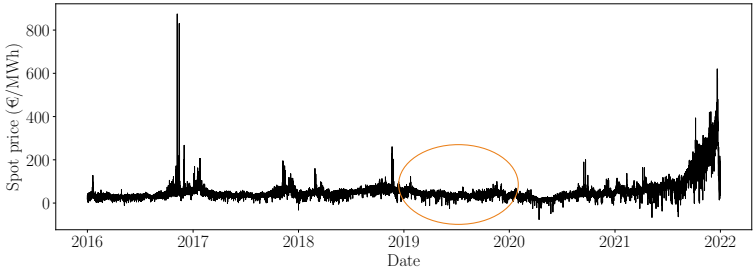

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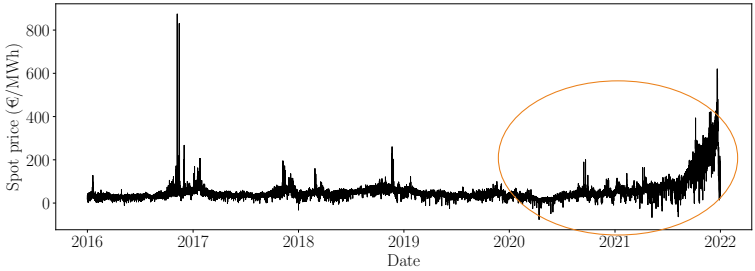
- 2020 and 2021: `AgACI` fails to ensure validity, and the various forecasting models considered[1] behave differently.



[1] Quantile Random Forests, Quantile Generalized Additive Models, Quantile Gradient Boosting, etc.

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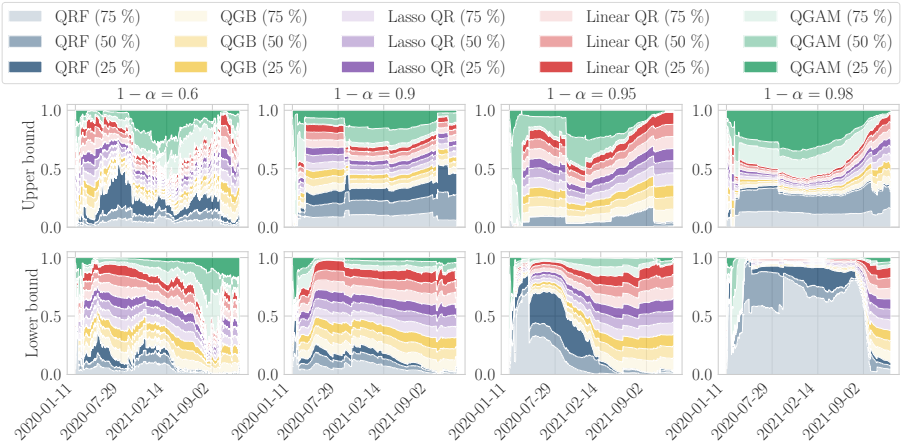

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↪ Weaken the objective and consider a more practical theoretical aim?

Avoiding data splitting: full conformal and out-of-bags approaches

- Full Conformal Prediction
- Jackknife+

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6. For a new point X_{n+1}, return

$$\widehat{C}_\alpha(X_{n+1}):=R_{q_{1-\alpha}(\mathcal{R})}\left(x;\hat{A}\right)=\{y\in\mathcal{Y}\text{ such that }\hat{r}\left(x,y\right)\leq q_{1-\alpha}\left(\mathcal{R}\right)\}$$

Some examples of nested sets (Gupta et al., 2022)

Example (Nested sets for the absolute value of the mean-regression residuals)**.**

$$s(x, y; \hat{\mu}) = |y - \hat{\mu}(x)| \iff \begin{cases} R_t(\cdot; \hat{\mu}) \equiv [\hat{\mu}(\cdot) \pm t] \\ \mathcal{T} = \mathbb{R}_+ \end{cases}$$

Some examples of nested sets (Gupta et al., 2022)

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$$s(x, y; \hat{\mu}) = |y - \hat{\mu}(x)| \iff \begin{cases} R_t(\cdot; \hat{\mu}) \equiv [\hat{\mu}(\cdot) \pm t] \\ \mathcal{T} = \mathbb{R}_+ \end{cases}$$

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Example (Nested sets for CQR).

$$\begin{aligned} & s\left(x, y; (\widehat{\mathrm{QR}}_{\text{lower}}, \widehat{\mathrm{QR}}_{\text{upper}})\right) \\ &= \max\left(\widehat{\mathrm{QR}}_{\text{lower}}(x) - y, y - \widehat{\mathrm{QR}}_{\text{upper}}(x)\right) \end{aligned} \iff \begin{cases} R_t\left(\cdot; (\widehat{\mathrm{QR}}_{\text{lower}}, \widehat{\mathrm{QR}}_{\text{upper}})\right) \\ \equiv \left[\widehat{\mathrm{QR}}_{\text{lower}}(\cdot) - t; \widehat{\mathrm{QR}}_{\text{upper}}(\cdot) + t\right] \\ \mathcal{T} = \mathbb{R} \end{cases}$$

Avoiding data splitting: full conformal and out-of-bags approaches

Full Conformal Prediction

Jackknife+

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SCP suffers from data splitting:

- lower statistical efficiency (lower model accuracy and higher predictive set size)
- higher statistical variability

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Can we avoid splitting the data set?

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✗ $\hat{A}$ obtained w. the training set $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ but not X_{n+1}.

Example ("Naive Idea" sets with an interpolating algorithm).

Assume $\mathcal{A}$ interpolates:

- $\hat{A} = \mathcal{A}((x_1, y_1), \ldots, (x_n, y_n))$
- $\hat{A}(x_k) - y_k = 0$ for any $k \in [\![1, n]\!]$

$\Rightarrow$ Naive method above *(with MAE score functions)* outputs $\{\hat{A}(X_{n+1})\}$ (a single point) for any new test point!

Full Conformal Prediction[9] does not discard training points!

- Full *(or transductive)* Conformal Prediction
  - avoids data splitting

[9] Vovk et al. (2005), *Algorithmic Learning in a Random World*

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- **Idea**: the most probable labels Y_{n+1} live in $\mathcal{Y}$, and have a low enough conformity score. By looping over all possible $y \in \mathcal{Y}$, the ones leading to the smallest conformity scores will be found.

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Full Conformal Prediction (CP): recovering exchangeability

For any candidate (X_{n+1}, y):

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$$\mathcal{S}_y^{(\text{train})} = \left\{ \mathrm{s}\left(\hat{A}_y(X_i), Y_i\right) \right\}_{i=1}^{n} \cup \left\{ \mathrm{s}\left(\hat{A}_y(X_{n+1}), y\right) \right\}$$

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- ✓ Test point treated in the same way than train points
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- ✗ Computationally costly

Definition (Symmetrical algorithm)**.**

A deterministic algorithm $\mathcal{A} : (U_1, \ldots, U_n) \mapsto \hat{A}$ is symmetric if for any permutation σ of $[\![1, n]\!]$: $\mathcal{A}(U_1, \ldots, U_n) \overset{\text{a.s.}}{=} \mathcal{A}\left(U_{\sigma(1)}, \ldots, U_{\sigma(n)}\right)$.

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Lemma (Exchangeable scores)**.**

If the algorithm $\mathcal{A} : (U_1, \ldots, U_n) \mapsto \hat{A}$ is symmetric, and $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable, then $S_1, \ldots, S_{n+1}$ are exchangeable, with

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Moreover

$$\begin{aligned} Y_{n+1} \in \widehat{C}_\alpha^{\text{Full}}(X_{n+1}) &:= \left\{ y \text{ such that } \mathrm{s}\left(\hat{A}_y\left(X_{n+1}\right), y\right) \leq q_{1-\alpha}\left(\mathcal{S}_y^{(\text{train})}\right) \right\} \\ &\Leftrightarrow \mathrm{s}\left(\hat{A}_{Y_{n+1}}\left(X_{n+1}\right), Y_{n+1}\right) \leq q_{1-\alpha}\left(\mathcal{S}_{Y_{n+1}}^{(\text{train})}\right) \\ &\Leftrightarrow S_{n+1} \leq q_{1-\alpha}(S_1, \ldots, S_n, S_{n+1}) \text{ !} \end{aligned}$$

Full CP: theoretical guarantees

Full CP enjoys finite sample guarantees proved in Vovk et al. (2005).

Theorem (Marginal validity of Full CP Vovk et al. (2005)).

Suppose that

(i) $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable,

(ii) the algorithm $\mathcal{A}$ is symmetric.

Full CP applied on $(X_i, Y_i)_{i=1}^{n} \cup \{X_{n+1}\}$ outputs $\widehat{C}_\alpha(\cdot)$ such that:

$$\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha(X_{n+1})\right\} \geq 1 - \alpha.$$

Additionally, if the scores are a.s. distinct:

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✗ Marginal coverage: $\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha(X_{n+1}) \,|\, X_{n+1} = x\right\} \geq 1 - \alpha$

Interpolation regime

Example (FCP sets with an interpolating algorithm)**.**

Assume $\mathcal{A}$ interpolates:

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$\Rightarrow$ Full Conformal Prediction *(with standard score functions)* outputs $\mathcal{Y}$ (the whole label space) for any new test point!

Split Conformal Prediction is a special case of Full Conformal Prediction

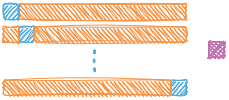
- Set $\hat{A}_y \equiv \hat{A}$, constant, independent of $\{(X_1, Y_1), \ldots, (X_n, Y_n)\} \cup \{(X_{n+1}, y)\}$
- Then, running Full Conformal Prediction corresponds to Split Conformal Prediction with $\#\mathrm{Cal} = n$.

Avoiding data splitting: full conformal and out-of-bags approaches

Full Conformal Prediction

Jackknife+

Jackknife: the naive idea does not enjoy valid coverage



- Based on leave-one-out (LOO) residuals
- $\mathcal{D}_n = \{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ training data
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Warning

No guarantee on the prediction of $\hat{A}$ with scores based on $(\hat{A}_{-i})_i$, without assuming a form of **stability** on $\mathcal{A}$.

Jackknife+ (Barber et al., 2021)

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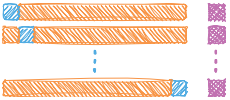

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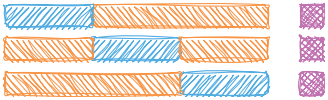

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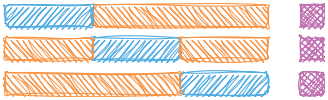

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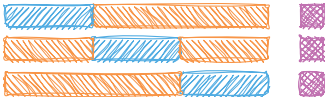

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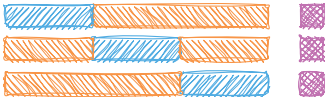
Recall $q_{\beta,\text{inf}}(X_1, \ldots, X_n) := \lfloor \beta \times n \rfloor$ smallest value of $(X_1, \ldots, X_n)$

CV+ Barber et al. (2021) (see also cross-conformal predictors: Vovk, 2015)



- Based on cross-validation residuals
- $\mathcal{D}_n = \{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ training data
- Split $\mathcal{D}_n$ into K folds $F_1, \ldots, F_K$
- Get $\hat{A}_{-F_k}$ by training $\mathcal{A}$ on $\mathcal{D}_n \setminus F_k$
- Cross-val predictions / predictive intervals

$$\mathcal{S}_{\text{up/down}} = \left\{ \left\{ \hat{A}_{-F_k}(X_{n+1}) \pm |\hat{A}_{-F_k}(X_i) - Y_i| \right\}_{i \in F_k} \right\}_k \cup \{\pm\infty\}$$

(in standard mean regression)

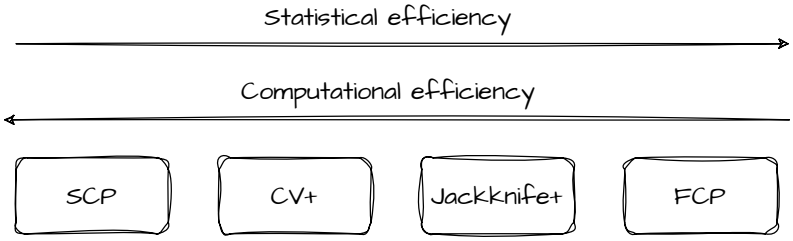
- Build the predictive interval: $[q_{\alpha,\text{inf}}(\mathcal{S}_{\text{down}}); q_{1-\alpha}(\mathcal{S}_{\text{up}})]$

Theorem (Marginal validity of CV+ Barber et al. (2021)).

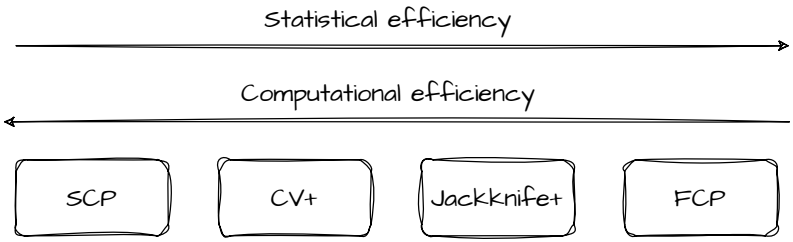
If $\mathcal{D}_n \cup (X_{n+1}, Y_{n+1})$ are exchangeable and $\mathcal{A}$ is symmetric: $\mathbb{P}(Y_{n+1} \in \widehat{C}_\alpha(X_{n+1})) \geq 1 - 2\alpha - \min\left(\frac{2(1-1/K)}{n/K+1}, \frac{1-K/n}{K+1}\right) \geq 1 - 2\alpha - \sqrt{2/n}$.

Recall $q_{\beta,\text{inf}}(X_1, \ldots, X_n) := \lfloor \beta \times n \rfloor$ smallest value of $(X_1, \ldots, X_n)$

General overview

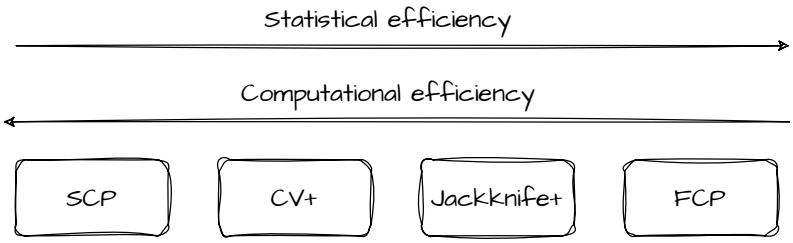


General overview



- Generalized framework encapsulating out-of-sample methods: Nested CP (Gupta et al., 2022) $\rightarrow$ extends *JK+/CV+ for any score.*

General overview



- Generalized framework encapsulating out-of-sample methods: Nested CP (Gupta et al., 2022) $\rightarrow$ extends *JK+/CV+ for any score.*
- Accelerating FCP: Nouretdinov et al. (2001); Lei (2019); Ndiaye and Takeuchi (2019); Cherubin et al. (2021); Ndiaye and Takeuchi (2022); Ndiaye (2022)

Non exhaustive references.

Angelopoulos, A. N., Candès, E. J., and Tibshirani, R. J. (2023). Conformal pid control for time series prediction. arXiv: 2307.16895.

Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2021). Predictive inference with the jackknife+. *The Annals of Statistics*, 49(1).

Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2022). Conformal prediction beyond exchangeability. To appear in *Annals of Statistics (2023)*.

Bastani, O., Gupta, V., Jung, C., Noarov, G., Ramalingam, R., and Roth, A. (2022). Practical adversarial multivalid conformal prediction. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc.

Bhatnagar, A., Wang, H., Xiong, C., and Bai, Y. (2023). Improved online conformal prediction via strongly adaptive online learning. In *Proceedings of the 40th International Conference on Machine Learning*. PMLR.

References ii

Cauchois, M., Gupta, S., Ali, A., and Duchi, J. C. (2020). Robust Validation: Confident Predictions Even When Distributions Shift. arXiv: 2008.04267.

Chernozhukov, V., Wüthrich, K., and Yinchu, Z. (2018). Exact and Robust Conformal Inference Methods for Predictive Machine Learning with Dependent Data. In *Conference On Learning Theory*. PMLR.

Cherubin, G., Chatzikokolakis, K., and Jaggi, M. (2021). Exact optimization of conformal predictors via incremental and decremental learning. In *Proceedings of the 38th International Conference on Machine Learning*. PMLR.

Dutot, G., Zaffran, M., Féron, O., and Goude, Y. (2024). Adaptive probabilistic forecasting of french electricity spot prices.

Gibbs, I. and Candès, E. (2021). Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc.

Gibbs, I. and Candès, E. (2022). Conformal inference for online prediction with arbitrary distribution shifts. arXiv: 2208.08401.

Gupta, C., Kuchibhotla, A. K., and Ramdas, A. (2022). Nested conformal prediction and quantile out-of-bag ensemble methods. *Pattern Recognition*, 127.

Lei, J. (2019). Fast exact conformalization of the lasso using piecewise linear homotopy. *Biometrika*, 106(4).

Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R. J., and Wasserman, L. (2018). Distribution-Free Predictive Inference for Regression. *Journal of the American Statistical Association*.

Ndiaye, E. (2022). Stable conformal prediction sets. In *Proceedings of the 39th International Conference on Machine Learning*. PMLR.

Ndiaye, E. and Takeuchi, I. (2019). Computing full conformal prediction set with approximate homotopy. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc.

Ndiaye, E. and Takeuchi, I. (2022). Root-finding approaches for computing conformal prediction set. *Machine Learning*, 112(1).

Nouretdinov, I., Melluish, T., and Vovk, V. (2001). Ridge regression confidence machine. In *Proceedings of the 18th International Conference on Machine Learning*.

Podkopaev, A. and Ramdas, A. (2021). Distribution-free uncertainty quantification for classification under label shift. In *Proceedings of the Thirty-Seventh Conference on Uncertainty in Artificial Intelligence*. PMLR.

Romano, Y., Patterson, E., and Candès, E. (2019). Conformalized Quantile Regression. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc.

Tibshirani, R. J., Barber, R. F., Candes, E., and Ramdas, A. (2019). Conformal Prediction Under Covariate Shift. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc.

Vovk, V. (2015). Cross-conformal predictors. *Annals of Mathematics and Artificial Intelligence*, 74(1-2).

Vovk, V., Gammerman, A., and Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer US.

Zaffran, M., Féron, O., Goude, Y., Josse, J., and Dieuleveut, A. (2022). Adaptive conformal predictions for time series. In *Proceedings of the 39th International Conference on Machine Learning*. PMLR.

https://conformalpredictionintro.github.io